Building Valid Arguments: An Introduction to Propositional Logic

*An original lesson plan inspired by **Ephraim Glick**'s Logic I (24.241) (MIT OpenCourseWare). Not affiliated with or endorsed by MIT OpenCourseWare or Ephraim Glick. Explore the original course: https://ocw.mit.edu/courses/24-241-logic-i-fall-2009/*

GRADE LEVEL	DURATION	TOPICS
Grades 9-12 (adaptable to intro college)	50 minutes	Logic

> *"What actually makes an argument logically valid, and how can we test it systematically rather than relying on gut feeling?"*

Learning Objectives

By the end of this lesson, students will be able to:

- Translate simple English arguments into propositional logic notation using the connectives AND, OR, NOT, and IF...THEN.
- Construct a truth table to test whether a given argument form is valid.
- Distinguish an argument's validity (does the conclusion follow from the premises?) from its soundness (are the premises actually true?).

Materials Needed

- Whiteboard with truth-table grid template pre-drawn
- Handout with 6–8 short English arguments of varying validity/soundness for translation practice
- Index cards, one connective symbol on each ($\wedge$ $\vee$ $\neg$ $\rightarrow$), for a quick matching warm-up

Lesson Procedure

Hook: A Valid Argument That Feels Wrong

8 min

Present this argument on the board: "All fish live in trees. Sharks are fish. Therefore, sharks live in trees." Ask students: is this a good argument? Most will immediately say no because the conclusion is false — use this to introduce the day's central distinction: this argument is actually valid (the conclusion really does follow from the premises), it's just not sound, because the first premise is false. Validity and truth are two different questions.

Direct Instruction: Connectives and Truth Tables

15 min

Introduce the four basic connectives (negation, conjunction, disjunction, the conditional) and how each is defined by a truth table showing every combination of true/false inputs. Work through one complete example on the board translating an English sentence (“if it rains, then the game is cancelled”) into symbolic notation and building its truth table row by row.

Guided Practice: Translate and Test

18 min

In pairs, students take 3–4 arguments from the handout, translate each into symbolic notation, and build a truth table to check whether the argument form is valid (there is no row where all premises are true and the conclusion is false). Circulate to catch common translation errors, especially around “if...then” and “unless.”

Closure: Validity vs. Soundness Sort

9 min

Present 4 short arguments on the board. As a class, quickly sort each into one of four boxes: valid and sound, valid but unsound, invalid (regardless of whether the premises happen to be true). Use this to reinforce that validity is about logical structure alone, entirely separate from whether the premises are actually true.

Discussion Questions

1. Can an argument be valid even if every single statement in it — premises and conclusion — is false? Try to construct one.
2. Why might it be useful to separate the question “is this logically valid?” from the question “are the premises true?” when someone tries to persuade you of something?
3. Truth tables get very large very quickly as you add more variables. Can you think of a faster way to check validity, even informally?
4. Do people in everyday arguments usually confuse validity with soundness? Can you think of a real example?

Assessment

Give students two new arguments they haven't seen before. They must translate each into symbolic notation, build a truth table, and state whether it is valid, and separately whether it is sound, with a one-sentence justification for each judgment.

Extension Activity

Challenge students to construct their own deliberately silly but logically valid argument (false premises, true-following conclusion) and swap with a partner to verify each other's truth table.